

Project: RAA-Gatekeeper

"The Birth of RAA: Establishing the .DLL Standard for AI Security"

Status: Development Phase 2 (Performance & Scaling) **Origins:** Concept established April 5, 2026 (Ref: *The Birth of RAA* PDF) **Intelligence:** xAI Grok-1.5 / Google Gemini (Forensic Oracles)

1. The RAA Philosophy: Modular Trust

Restrictive Ability (RAA) is a recursive framework designed to turn "Restrictive Ability" into the standardized, modular equivalent of **Dynamic Link Libraries (.dll)** for the AI ecosystem.

The READ-ONLY AUDITOR (ROA)

RAA-Gatekeeper serves as the primary **Read-Only Auditor**. It acts as a non-invasive, forensic layer between local codebases and AI Agents.

- Zero Mutation:** The software never edits or sanitizes files; it preserves forensic integrity.
- Standardized Modular Trust:** Like a .dll provides a specific, trusted function to a program, an RAA-certified file provides a specific, trusted context to an AI.

2. Key Modules & Forensic Capabilities

A. The Forensic Gatekeeper (Rust/Tauri)

- DNA Fingerprinting:** Uses mathematical hashes to ensure that "what was audited is what is being passed."
- Junk Filtering:** Surgically excludes macOS metadata (`__MACOSX, ._`) to optimize LLM oracle efficiency.

B. LLM Security Oracle

- Signature Detection:** Identifies "Living-off-the-Land" patterns (e.g., `curl | bash`).
- Compliance Checks:** Audits against NIST/OWASP standards (e.g., plaintext `.env` keys).
- Agent Hijack Guard:** Detects hidden instructions designed to subvert AI intent.

C. RAA Shell-Check

A pre-execution audit layer that filters terminal commands for destructive wildcards (`rm -rf`) and unauthorized `sudo` elevation.

3. The RAA Standard: "Trust, then Certify"

The ultimate goal is to move the AI industry toward **.raa Certification**.

- **The Ledger:** Every file interaction is recorded in a tamper-proof `.raa` forensic report.
- **The Vision:** To build an ecosystem where "Restrictive Ability" is a sellable, auditable, and lucrative standard—providing the trust infrastructure the AI revolution currently lacks.